

An Advanced Real-Time Facial Recognition System with People Counting, Video Search, and Blacklist-Based Access Control

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Abstract— This paper presents an integrated facial recognition platform that combines deep-learning based person detection with embedding-based face identification to deliver five operational modes within a single web interface: live camera recognition with unique-person counting, target search through a webcam stream, static image analysis, target search inside pre-recorded videos, and a security mode based on a persistent blacklist. The system relies on the YOLOv8 object detector to count people in the scene and on the dlib-based face recognition library to compute 128-dimensional facial embeddings used for identification. A Streamlit front-end orchestrates the modules and provides interactive controls, while pyttsx3 produces non-blocking French voice alerts when a target is detected, when a blacklisted person appears, or when the configured occupancy limit is exceeded. Experimental observation on a standard webcam (640×480) shows that the proposed pipeline operates at an interactive frame rate, correctly identifies enrolled subjects with a Euclidean tolerance of 0.5, and reliably triggers audio-visual alerts. The contribution is a reproducible, modular and low-cost framework suitable for small-scale access control, classroom attendance, and event monitoring scenarios.

Keywords— facial recognition, YOLOv8, deep learning, people counting, blacklist, access control, Streamlit, computer vision.

I. INTRODUCTION

Automated facial recognition has become one of the most visible applications of computer vision, powering use cases that range from smartphone unlocking to large-scale video surveillance. Recent progress in convolutional neural networks and metric-learning embeddings has made it possible to identify individuals from a single reference image with high accuracy, while general-purpose object detectors such as the YOLO family allow simultaneous counting and localization of people in real time [1], [2].

Despite this progress, building an end-to-end system that is simultaneously accurate, interactive and easy to deploy remains challenging for small organizations. Most open-source demonstrations focus on a single feature—either

detection, or identification, or counting—and rarely combine the audio feedback and access-control logic required by real operational settings such as classrooms, laboratories or event entrances.

This paper describes the design and implementation of a unified facial recognition platform that integrates five complementary modes inside a single Streamlit web application: (i) live camera recognition with unique-person counting, (ii) live target search from a user-supplied photograph, (iii) static image analysis, (iv) target search inside pre-recorded videos, and (v) a security mode based on a persistent blacklist with audible alerts. The remainder of the paper is organized as follows. Section II reviews related work. Section III presents the methodology and system architecture. Section IV reports the experimental results and discusses the behavior of each module. Section V concludes and outlines future improvements.

II. RELATED WORK

Modern face recognition pipelines typically combine three stages: face detection, alignment, and embedding-based identification. The *face_recognition* library [3], built on top of dlib, provides a widely used implementation of a ResNet-based encoder that maps each detected face to a 128-dimensional vector, enabling identification through Euclidean distance comparison.

For people detection and counting, the YOLO family of single-stage detectors has become the de-facto standard thanks to its favorable accuracy/speed trade-off. YOLOv8 [2] further improves anchor-free detection and provides lightweight variants such as *yolov8n* suitable for CPU inference. Several works combine YOLO-based detection with face embeddings for crowd analytics or attendance management, but few of them expose an interactive interface or integrate access-control rules and voice feedback in a single tool. The system described here addresses this gap by unifying these components in a modular Python application.

III. METHODOLOGY

A. System Overview

The proposed application is implemented in Python and structured as a single Streamlit script that exposes a sidebar menu with six entries: home, camera mode, image-based search, image analysis, video search, and security mode. Each entry triggers a dedicated processing pipeline while sharing common resources such as the reference encodings and the session state. Figure 1 (conceptual) illustrates the global architecture: the front-end captures user inputs and frames, dispatches them to the appropriate computer-vision module, and renders annotated outputs together with statistics and voice alerts.

B. Face Enrollment and Encoding

Reference images of authorized people are stored in the directory *personnes_reference/*. File names follow the convention *Name_k.jpg*, where the optional suffix *_k* allows several photographs of the same person to be enrolled. A regular expression normalizes the file name to extract the canonical identity. For every reference image, the system loads the picture, detects the face, and computes the 128-dimensional embedding produced by the dlib ResNet model. The set of pairs (encoding, name) is cached using *st.cache_data* to avoid recomputation between Streamlit reruns.

C. Identification and Counting Pipeline

In camera mode, each captured frame is processed by two parallel branches. The first branch runs YOLOv8-nano with an input size of 320 pixels and a confidence threshold of 0.5, keeping only detections of class *person*. The second branch resizes the frame by a factor of 0.5 to accelerate face localization, then computes embeddings for every detected face. Each embedding is compared against the gallery of reference encodings using the Euclidean distance, with a recognition tolerance of 0.5. The smallest distance below the threshold determines the identity; otherwise, the face is labeled as “unknown” and a stable hash of the embedding is used as a pseudo-identifier so that the same unknown person is not counted twice within a session. A set of already-counted identities is maintained in the Streamlit session state, allowing the system to report the number of unique persons observed since the last reset.

D. Image and Video Target Search

In image analysis mode, the user uploads a single picture which is decoded with OpenCV, converted to RGB, and processed by the same recognition pipeline as the camera mode. Recognized faces are framed in green, unknown faces in red, and aggregated statistics are displayed alongside the annotated image.

In image-based search mode, the user provides a reference photograph of a target person and a name. The encoding extracted from this photograph becomes the only template used to compare against faces detected in the live webcam feed; a positive match triggers a green bounding box, a confidence score, and a French voice alert that is rate-limited to one utterance every five seconds. Video-based search follows the same logic but operates on an uploaded video file, sampling one frame every quarter of a second to keep the analysis tractable while preserving temporal coverage.

E. Security Mode and Voice Alerts

The security mode introduces a persistent blacklist stored in the directory *blacklist/*. Operators can add or remove entries through the interface; each addition recomputes the embedding of the new image and updates the in-memory list. During camera operation, every detected face is compared not only against the white-list of authorized people but also against the blacklist using a tolerance of 0.5. A positive blacklist match displays a red overlay banner, marks the bounding box as *INTERDIT*, and triggers a spoken warning, throttled to one alert every eight seconds to avoid spam.

All audio messages are produced by *pyttsx3* in a separate daemon thread. On Windows, the COM library is initialized in the worker thread to ensure compatibility with the Streamlit runtime. A French voice is selected when available. An additional rule monitors the YOLO person count: whenever the number of simultaneously visible persons exceeds a configurable limit (default 1), a voice alert announces the over-occupancy event, again rate-limited to one notification every ten seconds.

IV. RESULTS AND DISCUSSION

A. Experimental Setup

The system was evaluated on a standard laptop equipped with an integrated webcam at 640×480 resolution. The YOLOv8-nano weights (*yolov8n.pt*) were used without retraining. The reference gallery contained several enrolled identities, each represented by one to three photographs. The recognition tolerance and YOLO confidence threshold were both fixed to 0.5, values commonly recommended in the literature [2], [3].

B. Qualitative Results

In camera mode, the system correctly drew green bounding boxes and identity labels around enrolled subjects, while unknown faces were marked in orange and counted only once per session through the embedding-hash mechanism. The combination of YOLO and *face_recognition* allowed the system to display, in real time, the number of detected persons, the number of unique identities since startup, and the elapsed monitoring duration.

In image-based and video-based search modes, the target person was reliably localized whenever a sufficiently frontal face was visible. The textual status (“FOUND” / “SEARCHING”) and the voice notification provided clear feedback to the operator. Video search additionally reported the time stamp of the first match, expressed in minutes and seconds.

In security mode, adding a person to the blacklist produced an immediate effect on the next camera frame: the matching face was framed in red, the badge “INTERDIT” was displayed, a translucent red banner appeared on top of the frame, and a French spoken warning was emitted.

Conversely, persons enrolled in the white-list but absent from the blacklist were displayed in green, demonstrating the correct interaction between the two galleries.

C. Discussion

The experiments confirm that combining a lightweight detector with a strong embedding-based recognizer is sufficient to build a usable real-time identification system on commodity hardware. The use of frame downsampling by a factor of two and YOLO at 320-pixel input size keeps the per-frame latency low enough for an interactive Streamlit interface. The session-state mechanism prevents double counting of the same identity, which is critical for any attendance or occupancy use case.

Several limitations remain. First, recognition accuracy degrades for non-frontal faces, occlusions, or strong illumination changes, as is well known for embedding-based methods. Second, the system relies on a single reference embedding per identity by default; enrolling several photographs improves robustness but increases comparison cost. Third, the voice subsystem is platform-dependent and required explicit COM initialization on Windows. Finally, the blacklist comparison is performed independently from the white-list comparison, which may produce conflicting labels when the same person is enrolled in both galleries; a stricter precedence rule would resolve this.

V. CONCLUSION

This paper presented an integrated facial recognition platform that unifies live recognition, target search, image and video analysis, people counting, and blacklist-based access control inside a single Streamlit application. By combining the YOLOv8 detector with the dlib-based *face_recognition* library and a non-blocking text-to-speech subsystem, the proposed system delivers an interactive and reproducible solution suitable for small-scale security and monitoring scenarios. Future work will focus on improving robustness through multi-view enrollment, replacing the dlib encoder with a more recent transformer-based face model, persisting the security log to a database, and deploying the application as a containerized service so that it can be operated remotely and integrated with physical access-control hardware.

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GITHUB:

<https://github.com/BAIHA-Fatima-Ezzahrae/Face-Recognition-System-with-Real-Time-Detection-and-Access-Control>